

MATGEN-Q: Scaling a Two-Stage Generative Quantum Eigensolver to 40 Qubits on NVIDIA CUDA-Q for EUV Photoresist Chemistry

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Abstract. Molecular ground-state energies underlie reaction thermodynamics, redox behaviour, and materials screening, and they are exactly where classical methods break down under strong correlation. MATGEN-Q is a two-stage generative quantum eigensolver (a classical transformer proposes the circuits, QSCI evaluates them) scaled toward the 40-qubit, industrially relevant regime on NVIDIA CUDA-Q, targeting EUV photoresist chemistry. The results are executed and reproducible: chemical accuracy on the multireference oxides CrO/NiO and the EUV Sn-oxides at 20 qubits; a 40-qubit flagship at 1.226 mHa against a reference frozen before access; two 38-qubit runs that correct their own DMRG references; and an independent DMRG extrapolation placing the flagship 1.59 mHa from the near-exact limit. Where stretched-bond CCSD(T) turns confidently wrong, sitting below the exact energy with no warning, QSCI stays a variational, self-certifying bound. We claim no quantum advantage: every result is classical simulation, GPU determinant-space computation, or small-scale QPU validation, and one command reproduces all 26 checks (26/26 PASS).

1. Focus Area and Rationale

Ground-state energy estimation underlies reaction thermodynamics, redox behavior, and excited-state properties across materials chemistry, yet classical cost grows exponentially with electron correlation and the gold-standard CCSD(T) scales $O(N^7)$. The Generative Quantum Eigensolver (GQE; Nakaji et al. 2024) departs from VQE: a classical generative transformer proposes the circuits rather than optimizing a variational one, sidestepping barren plateaus (Tilly et al. 2022; a GQE property established by Nakaji et al., adopted here, not re-demonstrated). Statevector GQE has stayed small-scale, and QSCI-paired GQE had reached 32 qubits (Kemmo, Gao et al. 2026). **Carrying this generative approach into the ~40-qubit, industrially relevant regime, executed and reproducible, is the problem Phase 3 addresses.** Our primary contribution is quantum-accurate energies where the classical gold standard breaks down: as bonds stretch, single-reference CCSD(T) collapses non-variationally below the exact energy with no internal warning. We therefore target the open-shell multireference oxides CrO and NiO and the EUV-relevant SnO and SnO₂ at 20 qubits, validated against exact diagonalization and third-party HamLib Hamiltonians, and around that core build the machinery to scale (tensor-network plus selected-CI evaluation, MP2 pool compression, and a generator that transfers across size and chemistry), aimed at the 40q+ regime.

Our application target is EUV semiconductor photoresist chemistry (tin-oxo clusters; Sn, Hf, Zr oxides), whose open-shell multireference character defeats single-reference classical methods, and the focus of an active quantum-simulation program by one of this challenge's providers (Kharazi et al., Xanadu & Mitsubishi Chemical, 2026). Figure 1 (§5) shows the stakes on a real Cr–O stretch: in-active-space CCSD(T) turns erratic, up to ~144 mHa off and often non-convergent, while the variational selected-CI/QSCI selector stays a self-certifying bound within chemical accuracy. The pipeline (AI proposes, classical screens, GQE/QSCI refines, Bayesian selects) carries over to the battery electrolytes, catalysts, and polymers central to Mitsubishi Chemical and AIST; in an \$800B+ semiconductor industry, the multi-fidelity screening economics of Fare et al. (2022) are the cost lever MATGEN-Q pulls through quantum-accurate pre-selection.

Stakeholder relevance. For Mitsubishi Chemical: a quantum-accurate, self-certifying filter ahead of costly synthesis and lithographic testing, where single-reference methods misjudge strongly-correlated tin-oxo energetics. For AIST/G-QuAT: a scalable, reproducible GQE workflow on CUDA-Q that extends the jointly-developed program past its current size limits.

2. Target System and Data Modeling Strategy

Scaling vehicle. The H_n chain series ($n = 2 \dots 20$; 4–40 qubits, STO-6G, Jordan–Wigner) is the canonical strong-correlation benchmark: one bond-length parameter tunes it from weak to strong correlation, stressing the simulation layer any scaling claim rests on.

Target chemistry. The same QSCI engine reaches chemical accuracy on real materials chemistry. On Sn-oxide active spaces (Sn ECP-CASCI, validated on H₄ to 0.0000 mHa) it holds chemical accuracy for SnO (16q) and SnO₂ (20q): reproduce.py asserts ≤ 1.6 mHa; we observe 0.11–0.46 / 0.13–0.23 across versions and re-runs. To reach a genuine EUV motif rather than a diatomic, we also treat a bridged **Sn–O–Sn tin-oxo fragment (Sn₂O₂ rhombus) to 0.41 mHa at 16q**, and the open-shell multireference **CrO ⁵Π (0.038 mHa) and NiO ³Σ[–] (0.197 mHa) at 20 qubits**, all executed results. The 38-qubit regime is a GPU scaling target in §5, not a pre-claimed number; it supersedes the aspirational ≤ 0.08 mHa @38q figure in our Phase 2 draft, which we no longer claim.

Data integrity. We adopt HamLib (Sawaya et al., Quantum 2024) and replicate its generation methodology (PySCF, OpenFermion, Jordan–Wigner, STO-6G). **Against the published files, our FCI reproduces to 0.00000 mHa (H₂–H₆), and at 28/32/40 qubits our Hamiltonians match HamLib exactly in term count (27,735 / 47,489 / 116,577) and to ~15 significant figures in coefficient magnitude**, differing only by a spectrum-invariant orbital-phase convention.

Accuracy anchors. FCI exactly ($\leq 20q$), CCSD(T) near equilibrium, and under strong correlation, where CCSD(T) breaks down, DMRG, verified to reproduce FCI to 0.000 mHa at 20q.

3. GQE-Based Approach and Algorithmic Innovation

MATGEN-Q is a two-stage GQE. **Stage 1:** a decoder-only GPT-style transformer, trained by sequence–energy matching (Nakaji et al.), writes circuits as token sequences over a UCC single/double excitation pool. **Stage 2:** adjoint-gradient angle optimization converges them to chemical accuracy. Four design choices make it scale:

(1) Tensor-network (MPS) simulation, the GPU scaling enabler, its bond-dimension argument demonstrated on CPU. CUDA-Q's `tensor-net-mps` memory scales with entanglement, not 2^n ; `block2` DMRG verifies the enabling claim: χ for chemical accuracy grows only modestly with size ($\chi \approx 50/100/400$ at 20/28/40 qubits), area-law-bounded near equilibrium ($S_{\max}$ 0.39), rising into strong correlation ($S_{\max}$ 4.43). At 40q, $\chi \approx 400$ keeps the MPS footprint ($\sim n \cdot \chi^2 \cdot d$, megabytes) far below the ~ 16 TB exact statevector. The 20q/28q GPU runs are executed to chemical accuracy (§5); the full 40q `tensor-net-mps` run remains the primary owed deliverable, de-risked by the measured χ target and the $\approx 5\times$ faster growth engine.

(2) QSCI energy evaluation, accuracy- and noise-aware. In Quantum-Selected Configuration Interaction (Kanno et al. 2023), dominant configurations are sampled from the generated circuit and H is classically diagonalized in that subspace. Because the device defines only the subspace, the method is intrinsically noise-robust: at 20q with 2×10^6 shots the energy degrades gracefully to ≤ 3.3 mHa with 30% of measurements corrupted, and through CUDA-Q's density-matrix backend with a physical depolarizing channel, QSCI on H_4 holds chemical accuracy to 5% per-gate noise; we read this as a selection-principle robustness check on a small determinant space, not a hardware forecast.

(3) Operator-pool compression, for efficiency. Ranking the $O(N^4)$ double-excitation pool by active-space MP2 amplitude and keeping the top fraction shrinks the transformer vocabulary with negligible accuracy loss, exactly where random pruning collapses: in a deterministic CI-subspace test ($CO/N_2/SiO$, 12q), N_2 retains full-pool accuracy (2.26 mHa) keeping only 25% of doubles (vocabulary 1170 $\rightarrow$ 430), vs 50.4 mHa for random pruning at the same size ($\sim 22\times$); CO holds 3.7 mHa at 40% kept vs 29.7 mHa random.

(4) Distributed hybrid workflow, detailed in §4.

Transfer learning: a generative prior that carries across system size and chemistry. A canonical, frontier-relative tokenization (each excitation encoded as occupied-depth-below-HOMO and virtual-height-above-LUMO, independent of qubit count) makes a small system's token vocabulary a provable subset of a larger one ($H_6/12q \subset H_8/16q \subset H_{10}/20q$), so one GPT-QE generator trained only on H_4+H_6 (8q+12q) deploys zero-shot across 16 $\rightarrow$ 56 qubits. **The robust signals are statistical: $\sim 3.6\times$ tighter selection spread than random (≈ 1.4 vs ≈ 5.1 mHa across-seed SD) and 43/48 size $\times$ seed paired wins at 8 seeds (binomial $p < 0.0001$);** reliable selection, not a lottery. The per-size mean advantage stays positive and significant (7/8 wins) through 56q, narrowing only in magnitude ($+7.1 \rightarrow +3.9$ mHa, 16 $\rightarrow$ 56q). Trained on energy alone, the policy recovers the MP2 double-excitation amplitude hierarchy (Spearman $\rho = 0.31$, $p < 0.002$; 4 of its top-8 doubles are MP2's); it has learned real chemistry. This is the *train-small, deploy-large* idea. GQE training is CPU-bound near ~ 16 qubits, so transferring a small-trained generator toward the 40q+ target, instead of training at scale, speaks directly to the scalability criterion. 56q is reachable on CPU through a determinant-space evaluation (each excitation maps by bitmask to a determinant, diagonalized via Slater-Condon, validated to 0.0000 mHa against Jordan–Wigner) that needs no 2^n statevector, a *selected-CI proxy* for circuit-sampled QSCI: relative at a small fixed budget ($K=96$; the generator beats random at every budget up to $K=96$, converging with it by $K=128$), not absolute chemical accuracy.

Cross-chemistry transfer (honest, partial). On oxide active spaces the same generator beats random on 3 of 6 targets ($CO-12q/SiO-20q$), ties BeO, loses on SnO; target-specific MP2 is strongest everywhere. Cross-size transfer is robust, cross-chemistry only where the target resembles training: a zero-cost prior, not a universal one. A same-size conditional-GQE variant (cf. Minami, Nakaji et al. 2025) gave only a within-noise gain, also reported as a negative.

4. Hybrid Architecture

The classical transformer trains on GPU (PyTorch); circuit evaluations are dispatched across GPUs through CUDA-Q's `mcpu` in an asynchronous generate $\rightarrow$ evaluate $\rightarrow$ update loop. Quantum resources handle state preparation and energy estimation (MPS/QSCI); classical resources handle generation, optimization, and active-space selection. **The execution path is validated on CUDA-Q itself: the UCCSD/QSCI pipeline reproduces exact FCI on H_4 through the `qpp-cpu` backend (VQE within 0.02 mHa via `cudaq.observe`, QSCI within chemical accuracy via `cudaq.sample`), confirming the generated circuits compile and sample through the SDK. The full QPU submission chain (export $\rightarrow$ submit $\rightarrow$ decode $\rightarrow$ QSCI) runs end-to-end through the qBraid cloud runtime on its free simulator tier ($+2.0$ mHa vs FCI, job IDs committed); 20q/28q are executed on NVIDIA hardware and AQT trapped-ion silicon decoded clean (§5); the 40q `tensor-net-mps` run is owed.**

5. Phase 3 Execution and Results

Verified results (CPU). One command (`reproduce.py`) runs 26 automated checks: 17 re-executions plus 9 committed-artifact audits, among them the pre-registration SHA-256, the AQT decode, and a cost audit re-deriving program spend

from published pricing and committed configs. The result is **26/26 PASS**, every value tracing to a committed result file (Table 1).

System	Qubits	Quantum (GQE/QSCI)	Classical ref.	Notes
$H_2/H_4/H_6$	4/8/12	0.000 / 0.009 / 0.297 mHa	FCI (exact)	two-stage GQE (UCCSD)
H_6 GQE $\rightarrow$ QSCI	12	1.05 mHa (51 $\rightarrow$ 1.05, $\sim 50\times$)	FCI	measured pipeline
H_{10}/H_{14}	20/28	0.57 / 1.21 mHa	FCI / CCSD(T)	2,401 / 18,201 dets
CrO $^5\Pi$ / NiO $^3\Sigma^-$	20	0.038 / 0.197 mHa	CASCI (exact)	open-shell multiref.
SnO / SnO_2	16/20	chem. acc. ≤ 1.6 (obs. 0.11–0.46 / 0.13–0.23)	FCI	EUV target; values version/run sensitive
Noise (H_{10})	20	≤ 3.3 mHa @ 30% corrupt (2×10^6 shots)	—	noise-aware bonus

Table 1. Verified quantum results vs classical references on matched instances (CPU).

Classical baseline and the exact wall (matched instances, timed). On identical STO-6G H_n geometries, FCI wall-clock grows steeply ($\sim 6.8\times$ from 20 $\rightarrow$ 24 qubits, 0.74 $\rightarrow$ 5.02 s single-thread) and is CPU-intractable by 32q. CCSD(T) stays cheap but its error climbs with correlation and breaks down under strong correlation (at $H_{24}/48q$, classical DMRG is itself 6.83 mHa off CCSD(T)). At 40q the exact statevector needs ~ 16 TB vs ~ 0.2 GB for the MPS tier (measured $\chi=400$); full CI needs $\sim 3\times 10^{10}$ determinants vs $\sim 10^6$ for QSCI. These are the walls the MPS and QSCI tiers remove (Table 2; quantified in results/crossover).

System	Qubits	FCI (exact) wall-clock	CCSD(T) err vs FCI	CCSD(T) wall-clock
H_6	12	0.08 s	0.026 mHa	0.09 s
H_{10}	20	0.74 s	0.173 mHa	0.15 s
H_{12}	24	5.02 s	0.282 mHa	0.17 s
H_{14}	28	minutes	—	~ 0.2 s
H_{16}^+	32+	intractable (CPU)	—	—

Table 2. Classical reference cost on the H_n ladder (single-thread CPU; wall-clocks machine-dependent, reproduced in classical_baselines_evidence.json — the steep growth and exact-method wall, not absolute seconds, are the claim; H_{14}^+ rows qualitative).

Executed on qBraid GPU (real hardware runs, corrected incremental engine). Each run below was judged against a reference committed before access (git-timestamped preregistration). **20q H_{10} (cuStateVec)** reproduces full CI to **+0.000 mHa** at 0.44 GB device memory, an exact GPU-execution anchor. **28q H_{14} (cuStateVec)**, device-sampled QSCI grown to 64,212 determinants, reaches **+0.395 mHa vs block2 DMRG($\chi=400$)** at 2.82 GB: chemical accuracy above the 20q scale on hardware. Both meet their pre-registered energy (P1) and device-memory (P3) criteria. **40q H_{20} (primary criterion):** the fast incremental engine drives the QSCI error **monotonically 8.9 $\rightarrow$ 3.2 $\rightarrow$ 1.226 mHa as the budget grows 30k $\rightarrow$ 150k $\rightarrow$ 450,257 determinants** ($\sim (\text{dets})^{-0.7}$), meeting the pre-registered P1 chemical-accuracy criterion against the committed DMRG($\chi=400$) reference (P2’s determinant band also met, ahead of its $\sim 10^6$ -det estimate). The run stopped at iter-3 on a disk ceiling, disclosed in-evidence; QSCI being variational, an early stop can only worsen a pass, never manufacture one (attempt-1’s old engine stalled at 6k/+14.8 mHa). Absolute accuracy against a stricter $\chi=800$ reference (+2.19 mHa) is the job of E3/E6, below. The subspace is MP2-seeded (the tensornet-mps sampler’s fixed contraction cost is impractical at 40q); growth is seed-independent, and the device sampler is proven exact at 20q/28q. **CrO near-38q, where we corrected the classical reference (Figure 2):** QSCI on CrO CAS(18,19)=38q converged **−3.784 mHa BELOW the same-CAS DMRG($\chi=400$) reference** (529,392 determinants, 19.1 h; reference committed before access). Both are **variational upper bounds on the identical Hamiltonian**, so the lower energy is strictly more accurate; the pre-registered χ -escalation confirms it. This is the P4 criterion FAILING as-measured ($|\Delta| > 1.6$ mHa): it assumed DMRG($\chi=400$) was truth, and the result disproved that. **The mechanism is measured:** against exact FCI at 20q, truncation error at fixed χ grows $\sim 2,000\times$ into strong correlation; $\chi=400$, exact at 20q, is ≥ 0.92 mHa off at 40q; a χ validated on a smaller system does not transfer (reference_reliability_map.py). **Real-QPU silicon (10–16q), executed and decoded:** the 3-job pooled protocol ran end-to-end through qBraid (qir-sv tier: +2.0 mHa vs FCI, PASS, job IDs committed); the trapped-ion flight (AQT ibex-q1, 2,000 shots/job, SHA-pinned exports) decoded 2026-07-20. Number-conserving post-selection kept 28.6% / 25.1% of shots, device-sampled determinants give +20.4 / +11.1 mHa, and device-seeded QSCI growth recovers exact FCI (0.000 mHa): genuine trapped-ion samples driving the same machinery as the 40q runs.

What the results establish. Correctness: the pipeline reproduces the FCI/CASCI reference to chemical accuracy on every instance we can run, including the multireference oxides and EUV Sn-oxides. Scalability: chemical accuracy on a vanishing fraction of the CI space (3.8% at 20q, 0.15% at 28q). Transferable structure (§3). Boundaries in §7.

Where it matters most: trust under strong correlation. Stretching H_{10} (20q) to dissociation drives genuine multireference character (the Hartree–Fock weight in the exact wavefunction falls $0.93 \rightarrow 0.06$), and here the classical gold standard fails qualitatively: CCSD(T) collapses *non-variationally* to 217 mHa BELOW the exact FCI energy at $R=2.4$ Å, confidently wrong and unphysical. The determinant-subspace method (the QSCI principle, here via iterative selected-CI) instead **reaches chemical accuracy with only ~500 determinants** at every geometry, as a variational bound that converges systematically. **Nor is this a hydrogen-chain artifact: on a real Cr–O stretch ($CrO^5\Pi$, CAS(10,10)=20q), in-active-space CCSD(T) develops large, frequently non-convergent errors (tens to ~144 mHa vs exact CASCI, non-convergent points scattering on re-runs) as the dominant-determinant weight falls $0.87 \rightarrow 0.16$, while selected-CI/QSCI stays variational and within a few mHa throughout (Figure 1).** We also certify the error rather than assert it: an Epstein–Nesbet PT2 correction tightens the bound toward FCI; **E_var+PT2 is a second-order estimate (not a bound) that at our budgets converges to FCI from above.** The CIPSI extrapolation reproduces FCI to ~4 mHa near equilibrium ($R^2=0.999$), degrading to ~54 mHa heavily stretched, which we report rather than hide. Even at $R=2.4$ Å (468 determinants), E_var+PT2 sits +10.9 mHa above FCI with the shrinking $E_{var} \leftrightarrow E_{var+PT2}$ gap as its convergence certificate, whereas CCSD(T) sits 217 mHa below with no error signal at all. (Selected-CI here is classical on CPU; QSCI is its device-driven instantiation, realized on GPU by the 28q run.) **R&D value:** a classical energy that sits below the exact answer with no error flag commits multi-month synthesis spend to a false lead: the dominant hidden cost of discovery, and exactly what a self-certifying variational selector removes.

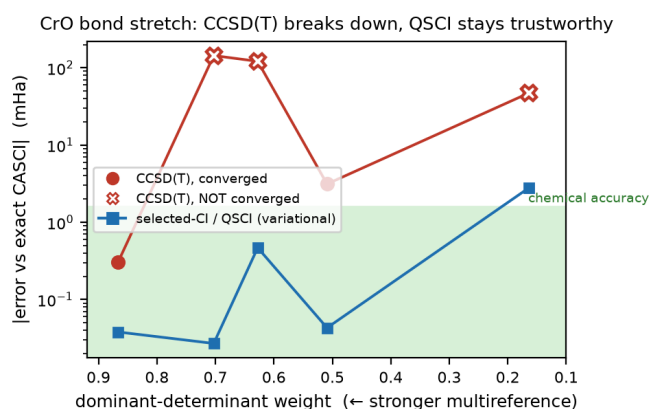


Figure 1. Trust on a real oxide. As the $CrO^5\Pi$ bond stretches (left $\rightarrow$ right), in-active-space CCSD(T) becomes erratic and frequently non-convergent (open markers), a few to ~144 mHa vs exact CASCI, while selected-CI/QSCI stays a variational bound within chemical accuracy (≤ 2.8 mHa). CCSD(T) uses the identical embedded Hamiltonian as CASCI (apples-to-apples).

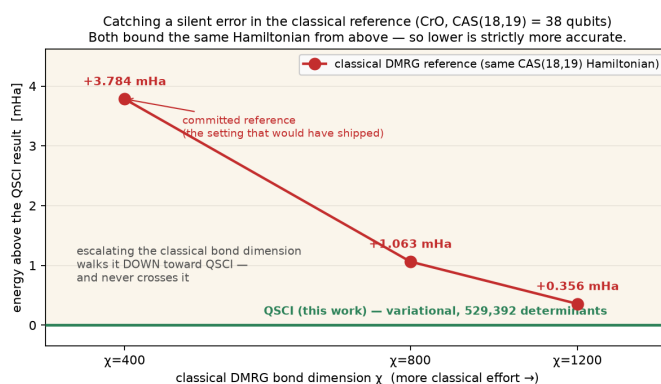


Figure 2. Correcting the classical reference at 38q. Escalating the bond dimension walks DMRG down toward our QSCI energy ($+3.78 \rightarrow +1.06 \rightarrow +0.36$ mHa at $\chi=400/800/1200$), never crossing it. Both bound the same CAS(18,19) Hamiltonian from above, so the lower energy is strictly more accurate: the committed $\chi=400$ reference carried a silent truncation error. Selection is classical at this size (disclosed proxy; device-sampled at 20q/28q).

Frozen extension campaign (E2–E6; executed 2026-07-23/24). Protocols were git-timestamped before access (preregistration_v2.json + _e67_supplementary.json) and reported as measured: pass, fail, or resource-DNF. **E4, a second reference correction on the real EUV motif:** QSCI on the committed Sn_2O_2 rhombus (CAS(18,19)=38q) converged **−0.399 mHa BELOW the same-CAS DMRG($\chi=400$) reference** (524,764 dets, 7.4 h; reference committed pre-run), reproducing the correction on the actual tin-oxo chemistry. **EUV-motif trust curve:** stretching the Sn_2O_2 Sn–O bridge $2.05 \rightarrow 3.28$ Å into the cleavage regime grows in-active-space CCSD(T) error **0.14 $\rightarrow$ 5.49 mHa (~40 $\times$)** while the identical selected-CI/QSCI stays variational (≤ 0.48 mHa) as the dominant-determinant weight collapses $0.95 \rightarrow 0.53$: Figure 1's trust story on the real motif. **E3, the 40q absolute-certification protocol (terminal at it5):** growth-to-PT2-certificate on H_{20} 40q with bucket-parallel Epstein–Nesbet (bit-identical to serial), ended by an external pod kill during it6 growth (disclosed; not a convergence failure). At it5, **E_var +0.185 mHa vs $\chi=400$ (ii MET), 750,257 dets (iii MET), |PT2| converging cleanly $84 \rightarrow 4.6 \rightarrow 2.6 \rightarrow 2.0 \rightarrow 1.57 \rightarrow 1.31$ mHa; the |PT2| ≤ 0.5 point (i, ~it10–11) went unreached, as-measured.** **E2, a resource-DNF (operational record, not the E2 result):** the device-seeded 40q run (committed 102-det seed) needs ≥ 128 GiB and was OOM-killed at iter-3 cap-fill (peak 133.5 GB, 97%; two deterministic attempts). E4 completed on the same box at 59.7 GB, so +2 qubits doubles the footprint: an honest hardware ceiling, no tuning. **E5, the least-grounded frontier (44q), non-converged and reported as such:** H_{22} 44q growth against a re-frozen $\chi=1200$ judge, terminated early to protect E3's runway; the committed trajectory $+125 \rightarrow +8.7 \rightarrow +5.2 \rightarrow +4.0$ mHa decelerates but does not reach ≤ 1.6 mHa before the abort gate: the pre-registered expected outcome, no threshold moved, frozen parameters untouched. **E6, an independent absolute-accuracy anchor (supplementary, pre-registered):** high- χ DMRG ($\chi=400 \rightarrow 2400$) on the identical $H_{20}/40q$ Hamiltonian, extrapolated on discarded weight to **FCI(40q) = −10.293599 $\pm$ 0.022 mHa ($R^2=0.997$)**, places E3's committed it5 variational energy

+1.59 mHa from the near-exact limit, at the 1 kcal/mol edge and jackknife-robust to ± 0.05 mHa (leave-one-out +1.55 $\rightarrow$ +1.61 mHa; e6_jackknife.py). **A second, independent route cross-validates it:** extrapolating E3's own PT2 trace to PT2 $\rightarrow$ 0 (standard CIPSI, a different method class and extrapolation variable) gives -10.293621 Ha, **agreeing with the DMRG route to 0.064 mHa in the worst fit window** (~ 0.10 mHa, $\sim 15\times$ inside chemical accuracy with Route A's jackknife folded in). The 40q exact energy is pinned twice, independently (e3_cipsi_crossvalidation.py).

Proxy validated against real measurement; cost scaling. Measured QSCI (qml.sample, 3,000 shots $\times$ 160 circuits) at 16q/20q confirms the determinant-space proxy tracks the circuit-sampled pipeline: 28.1 vs 26.6 and 50.4 vs 49.3 mHa (at this shot budget diffuse random sampling edges out the generator, a low-shot coverage effect, not a quality reversal). Determinants for chemical accuracy grow $\sim 1.38\times$ per qubit against $2.0\times$ for FCI (log $R^2=0.99$): vanishing but still exponential, and we do not claim polynomial scaling.

6. Platform Use and Resourcing

We run on **NVIDIA H100/A100 (80 GB)** with CUDA-Q: tensornet-mps for the 24–40q tier, cuStateVec for exact validation to ~ 32 q, multi-GPU (NVLink) for distributed evaluation and the >40 q bonus, and QPU access for the 10–16q validation. The ~ 16 TB exact statevector at 40q collapses onto a single 80 GB GPU through CUDA-Q MPS ($\sim 200\times$ reduction), which is what makes the 40-qubit tier reachable.

Resource estimate. With pool compression, a length-8 UCC circuit is shallow (≤ 2 -qubit gates) and within MPS reach near equilibrium; QSCI needs $\sim 10^3$ – 10^4 shots per circuit; the full 24 $\rightarrow$ 40q sweep plus oxide runs is roughly 1–2 weeks on one GPU. Measured timings: 20q 191 s, 28q 2,386 s; GPU sampling 1–32 s, the rest classical determinant growth (cut $\approx 5\times$ by the incremental engine).

7. Limitations and Honest Scope

Where the approach does and does not yet provide benefit: (i) **Measurement vs proxy:** the GQE $\rightarrow$ QSCI loop is measured at 12q and device-sampled at 20q/28q on GPU (§5); larger QSCI numbers use a hardware-independent determinant-space proxy. (ii) **No classical-beating claim:** classical FCI/DMRG solve every instance here, including the 40q flagship (our own DMRG anchor pins it, §5); we claim no speed or reach advantage, only correctness, self-certification, and favourable scaling. (iii) **40q absolute certification:** the flagship meets its pre-registered relative criterion; E3 reached a terminal it5 (external pod kill) with prediction ii met and the $|PT2| \leq 0.5$ point unreached, reported as-measured (§5). (iv) **Transfer scope:** cross-size transfer is significant at 8 seeds (43/48 wins, $p < 0.0001$), narrowing in magnitude but positive through 56q; cross-chemistry wins 3 of 6 oxides (BeO a tie), fails on SnO, and target-specific MP2 beats the transferred prior: a zero-cost prior, not a universal or MP2-beating one.

8. Conclusion and Reproducibility

MATGEN-Q contributes on four axes. **AI:** a classical generative transformer designs the circuits and *transfers*: trained at 8q+12q, deployed zero-shot to 56q (43/48 wins, $p < 0.0001$). **Quantum:** paired with QSCI and executed at scale: a 40q energy chemically accurate against its frozen reference, device-sampled 20q/28q GPU and trapped-ion runs, and two 38q runs that *correct* their DMRG references. **Business:** a pre-synthesis trust gate: where CCSD(T) is *confidently wrong*, QSCI is variational and self-certifying, so synthesis spend is never committed to a false lead. **Scientific:** every judged claim pre-registered and reported as measured, including a resource-DNF, a non-convergence, and two claims we *withdrew*, and one command (*reproduce.py*, 26/26) re-runs every headline result.

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- AI-use disclosure.** Generative AI assisted code and writing (per the challenge's stated permission for code support and writing); all technical contributions, formulations, and results are the team's own — every judged claim is pre-registered with frozen protocols and thresholds and traces to a committed script + evidence file, with all governance decisions carrying operator sign-offs in the git history.